

The VI Editor

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0.1 Introduction

The VI remains the goto editor for any xnix administrator. The editor can seem daunting at first and makes use of virtually every key on the keyboard. However, this is offset by the immense power it offers. It is a pure ASCII editor and does not create any meta text. It is also an excellent editor to code and offers a pure coding experience.

0.2 Working with vi

0.2.1 Invoking vi

You are most likely to invoke vi by using a command like

```
vi index.html
```

0.2.2 The three modes in vi

Command Mode - In this mode you only pass commands to process the text without actually entering text.

Input mode - In this mode you actually enter text. There are 10 keys which take us from the Command mode to the input mode. We press [Esc] to return to the Command mode from the input mode.

The last line or x mode - An internal shell for special or global commands e.g. saving text. You can reach this mode by pressing the [:] key while in the Command mode.

0.2.3 Command in the ex mode

0.2.4 Entering text

0.2.5 Recovering from a crash and escaping to the shell

We can usually recover quite well from a crash by restarting vi using the following command

```
vi -r <fname>
```

Table 1: List of commands in the ex mode

Command	Function
:w	save with current file name and continue editing
:w <i><fname></i>	save with new filename
:q	quit vi
:x	save and quit vi
:ZZ	save and quit vi

Table 2: List of commands in the Input mode

Command	Function
i, I	Insert text at current cursor location and start of the line.
a, A	Append text at current cursor location and to the end of the line.
o, O	open a new line for editing below and above the current line.
r <i><ch></i>	replace single character.
R	replace characters starting from current character to the right.
s	substitute the current character with multiple characters.
S	replace current line

We can temporarily escape to the shell from with vi using the `:sh` command.

0.2.6 Repeat factor

We can always repeat the next command n number of times by appending a repeat factor before the command. E.g. `20 i *` shall add twenty asterisks to the current line. Similarly, `3rP` will replace the next three characters with *P*.

0.2.7 Hopping around text

You can move to various points on the current line in the Command mode using the commands given in the table below:

There are also some interesting possibilities with the commands *B*, *E*, *W*. In, particular we can use the **G** command preceded by a number to move to a particular line number e.g. `4G` or `1G`. We can move to the last line simply by pressing **G**.

Table 3: List of in line navigation commands

Command	Function
0	move to the start of the line.
b	move to start of the current word.
e	move to the end of the current word.
w	start of next word.
\$	move to the end of the current line.

0.2.8 Deleting text

The commands to delete text are summarized in the table below:

Table 4: Commands to delete text

Command	Function
x and X	Delete the character at the cursor and before the cursor.
J	join the current line with the next line.
d\$ or D	deletes from cursor position to end of line.
dd	deletes the entire line.
dw	deletes the current word.
d⟨ <i>n</i> ⟩G	deletes from cursor position to line numbered <i>n</i> .

0.2.9 Yanking, Changing and pasting text

The commands to yank, change and paste text are summarized in the table below:

Table 5: Commands to copy, change and paste text

Command	Function
yy or Y	yank the current line.
y\$	yank from current cursor position to the end of the line.
yw	copies current word.
p	puts deleted or copied text to the right of the cursor or below the current line.
P	puts deleted or copied text to the right of the cursor or above the current line.
c	changes text.
~	changes case.

Do not forget that all the above commands will usually accept a repetition factor

0.2.10 Repeating the last command and undoing changes

We can repeat the last command executed by pressing the "." operator. While undoing the last change is done using the *u* command. You can undo all changes you have made in the current line by using the *U* command, but this only works as long as you have not left the current line.

0.2.11 Find and replace

Finding a word is accomplished using the `/word` command. Some other search commands are

Table 6: Searching through text

Command	Function
/pat	Search for first occurrence of pat
n	repeat last search in forward direction.
N	repeat last search in reverse direction.
fx	search for x in the current line.
Fx	search for x in reverse direction.

0.3 Configuring the editor

The ex mode offers a number of options for configuring the general working of the editor. However, these options will stay in effect only for the current session. To make invoke the options permanently we need to modify the `.exrc` file located in the `/home/yourusername` folder.

0.4 Assignment

1. Write a C program in vi to find the first four perfect numbers.
2. Write a program in C using vi to find a number in array using recursive binary search.